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4.1.2. Dictionary to dataframe

07:26

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame :

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

```
import pandas as pd

# Provided dictionary of lists
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
}

# Convert the dictionary to a DataFrame
df = pd.DataFrame(data)

# Display the original DataFrame
print("Original DataFrame:")
print(df)

# Adding a new row
new_name = input("New name: ")
new_age = int(input("New age: "))
df.loc[len(df)] = [new_name, new_age]

# Display the DataFrame after adding a new row
print("After adding a row:\n",df)

# Modifying a row
row_to_modify = int(input("Index of row to modify: "))
new_age_value = int(input("New age: "))
```

```
df.at[row_to_modify, "Age"] = new_age_value
```

```
# Display the DataFrame after modifying a row
```

```
print("After modifying a row:")
```

```
print(df)
```

```
# Deleting a row
```

```
row_to_delete = int(input("Index of row to delete: "))
```

```
df = df.drop(index=row_to_delete).reset_index(drop=True)
```

```
# Display the DataFrame after deleting a row
```

```
print("After deleting a row:")
```

```
print(df)
```

```
# Adding a new column
```

```
genders_input = input("Enter genders separated by space: ")
```

```
genders = genders_input.split()
```

```
# Ensure correct length
```

```
if len(genders) == len(df):
```

```
    df["Gender"] = genders
```

```
else:
```

```
    print("Error: Number of genders must match rows!")
```

```
    df["Gender"] = ["Unknown"] * len(df)
```

```
# Display the DataFrame after adding a new column  
print("After adding a new column:")  
print(df)
```

```
# Modifying a column  
df["Name"] = df["Name"].str.upper()
```

```
# Display the DataFrame after modifying a column  
print("After modifying a column:")  
print(df)
```

```
# Deleting a column  
df = df.drop(columns=["Age"])
```

```
# Display the DataFrame after deleting a column  
print("After deleting a column:")
```

print(df)

Explorer

Average time
0.043 s
43.00 ms

Maximum time
0.049 s
49.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

✓ Test case 1 49 ms Debug

Expected output

Actual output

Original DataFrame:

.....Name...Age

0....Alice...25

1....Bob...30

2..Charlie...35

New name: Susan

New age: 29

After adding a row:

.....Name...Age

0....Alice...25

1....Bob...30

2..Charlie...35

3....Susan...29

Index of row to modify: 0

New age: 27

After modifying a row:

.....Name...Age

0....Alice...27

1....Bob...30

2..Charlie...35

3....Susan...29

Index of row to delete: 3

After deleting a row:

.....Name...Age

0....Alice...27

1....Bob...30

Original DataFrame:

.....Name...Age

0....Alice...25

1....Bob...30

2..Charlie...35

New name: Susan

New age: 29

After adding a row:

.....Name...Age

0....Alice...25

1....Bob...30

2..Charlie...35

3....Susan...29

Index of row to modify: 0

New age: 27

After modifying a row:

.....Name...Age

0....Alice...27

1....Bob...30

2..Charlie...35

3....Susan...29

Index of row to delete: 3

After deleting a row:

.....Name...Age

0....Alice...27

1....Bob...30

Terminal

Test cases

Average time
0.043 s
43.00 ms



Maximum time
0.049 s
49.00 ms



✓ 1 out of 1 shown test case(s) passed
✓ 1 out of 1 hidden test case(s) passed



After modifying a row:

.....Name..Age

0...Alice...27

1...Bob...30

2..Charlie...35

3...Susan...29

Index of row to delete: 3

After deleting a row:

.....Name..Age

0...Alice...27

1...Bob...30

2..Charlie...35

Enter genders separated by space: F M M

After adding a new column:

.....Name..Age..Gender

0...Alice...27.....F

1...Bob...30.....M

2..Charlie...35.....M

After modifying a column:

.....Name..Age..Gender

0...ALICE...27.....F

1...BOB...30.....M

2..CHARLIE...35.....M

After deleting a column:

.....Name..Gender

0...ALICE.....F

1...BOB.....M

2..CHARLIE.....M

After modifying a row:

.....Name..Age

0...Alice...27

1...Bob...30

2..Charlie...35

3...Susan...29

Index of row to delete: 3

After deleting a row:

.....Name..Age

0...Alice...27

1...Bob...30

2..Charlie...35

Enter genders separated by space: F M N

After adding a new column:

.....Name..Age..Gender

0...Alice...27.....F

1...Bob...30.....M

2..Charlie...35.....M

After modifying a column:

.....Name..Age..Gender

0...ALICE...27.....F

1...BOB...30.....M

2..CHARLIE...35.....M

After deleting a column:

.....Name..Gender

0...ALICE.....F

1...BOB.....M

2..CHARLIE.....M

Terminal

Test cases